

CSE252:INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

L:2 T:0 P:2 Credits:3

Course Outcomes: Through this course students should be able to

CO1 :: understand the fundamental concepts of Artificial Intelligence, Machine Learning, intelligent agents, and their applications in robotics and autonomous systems.

CO2 :: apply preprocessing, feature engineering, and visualization techniques on datasets for developing AI models.

CO3 :: build supervised machine learning models for solving regression and classification problems using appropriate algorithms.

CO4 :: apply unsupervised learning techniques for clustering and dimensionality reduction to discover hidden patterns in data.

CO5 :: examine and improve machine learning models using appropriate performance metrics and validation techniques.

CO6 :: build simple AI-enabled solutions for real-world robotics and automation problems using Python and Scikit-learn.

Unit I

Foundations of Artificial Intelligence : Introduction to AI and its evolution, Types of Artificial Intelligence, AI vs ML vs Deep Learning vs Data Science, Problem-solving using AI and AI Development Lifecycle, Intelligent Agents and Agent Environment, Applications of AI in Robotics, Healthcare, Manufacturing and Smart Cities, Ethics in AI and Responsible AI, Introduction to Generative AI

Unit II

Data Preparation and Machine Learning Workflow : Introduction to NumPy, Pandas and Matplotlib, Understanding Data (Types of Datasets, Data Collection and Exploratory Data Analysis), Data Preprocessing (Handling missing values and Detecting outliers), Feature Engineering (Introduction to feature scaling, feature encoding and feature selection), Training and Validation (Train-Test Split and Cross Validation)

Unit III

Supervised Machine Learning : Regression (Linear Regression, Polynomial Regression, Decision Tree Regression), Classification (Logistic Regression, K-Nearest Neighbour, Introduction to Decision Tree Classifier and Random Forest, Naïve Bayes), Concepts of Bias, Variance, Underfitting and Overfitting

Unit IV

Unsupervised Learning : Clustering (K-Means Clustering, Hierarchical Clustering, Concept of DBSCAN), Dimensionality Reduction and Principal Component Analysis, Pattern Discovery (Association Rule Mining, Apriori Algorithm), Unsupervised Learning application in Robotics and Sensor data

Unit V

Model Evaluation and Improvement : Regression Metrics (MAE, MSE, RMSE, R^2 Score), Confusion Matrix and Classification Metrics (Accuracy, Precision, Recall, F1-score, ROC-AUC), Model Improvement (Cross Validation, Hyperparameter Tuning, Grid Search, Random Search), Introduction to Explainable AI

Unit VI

AI for Intelligent Systems and Robotics : Computer Vision (Overview of Image Classification, Overview of Object Detection) for Intelligent Systems and Robotics, Natural Language Processing (NLP Fundamentals, Chatbots and Conversational AI) for Intelligent Systems and Robotics, Reinforcement Learning (Concepts - Agent, State, Action, Reward) in context Intelligent Systems and Robotics, AI for Autonomous Robots and Future Trends in AI for Intelligent Systems and Robotics

List of Practicals / Experiments:

List of Practicals

- Installation and familiarization with Python, Jupyter Notebook and Google Colab.
- Data manipulation using NumPy and Pandas.

- Data visualization using Matplotlib.
- Exploratory Data Analysis on a real-world dataset – Titanic dataset and Student Performance dataset.
- Data preprocessing: handling missing values, encoding and feature scaling.
- Implementation of Linear Regression.
- Implementation of Logistic Regression.
- Classification using K-Nearest Neighbour.
- Classification using Decision Tree and Random Forest.
- Clustering using K-Means.
- Dimensionality Reduction using PCA.
- Performance evaluation using Confusion Matrix, Precision, Recall and F1-score.
- Hyperparameter tuning using Grid Search.
- Mini Project: Development of an AI solution for a robotics, automation or smart system application.

Text Books:

1. ARTIFICIAL INTELLIGENCE: A MODERN APPROACH by STUART RUSSELL, PETER NORVIG, PEARSON

References:

1. A FIRST COURSE IN ARTIFICIAL INTELLIGENCE by DEEPAK KHEMANI, MC GRAW HILL

2. DATA SCIENCE AND MACHINE LEARNING USING PYTHON by REEMA THAREJA, MC GRAW HILL

3. PYTHON MACHINE LEARNING BY EXAMPLE: UNLOCK MACHINE LEARNING BEST PRACTICES WITH REAL-WORLD USE CASES by YUXI (HAYDEN) LIU, PACKT PUBLISHING

4. FUNDAMENTALS OF MACHINE LEARNING: A BUSINESS PERSPECTIVE by PRATYUSH BANERJEE, SUPRITI MISHRA, SHIVASHANKAR CHARI, MC GRAW HILL

5. MACHINE LEARNING WITH PYTORCH AND SCIKIT-LEARN: DEVELOP MACHINE LEARNING AND DEEP LEARNING MODELS WITH PYTHON by SEBASTIAN RASCHKA, YUXI (HAYDEN) LIU, VAHID MIRJALILI, PACKT PUBLISHING